

LLM Semiotic Capacity:

measuring meaning communication

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Abstract

We introduce a model of Picca's semiotic channel principle. Semiotics studies how signs produce and communicate meaning. Channel capacity, from information theory, is the theoretical maximum rate at which data can be reliably transmitted over a communication channel, here a LLM used by AI agents.

Our semiotic channel communicating meaning is an implicit state machine LLM, where we operationalize meaning using a similarity function. In particular messages are interpreted by those that entail it, and we profile channel capacity by mapping message entropy against interpretation mutual information.

LLM that result in profiles with higher overall mutual information are sought for identification, certification and comparison of LLMs. This provides a rigorous and actionable framework for understanding, evaluating, and designing LLM-mediated communication between AI agents.

Synopsis

Consider a LLM as a communication channel between AI agents. We will train our LLM, a pair of state machines, in stages. First stage, state machine $Ecila(S, O')$ simulates a word embedding, where transition function S is an implicit recurrent deep learning model [7] generating states that are contextual representations of tokens, and whose output function O' then applies a feedforward network, resulting in key-value pairs and updated queries. Second stage, state machine $Alice(S, O)$ then uses this attention mechanism data to define another output function O . Each state has a distribution over fixed points from which we can choose the most likely token to output. $Alice(S, O)$ is trained to minimize perplexity regularized by the average Renyi α -entropy of the distributions.

$Alice(S, O)$ is a state machine LLM that combines an implicitRNN transition with an attention based output over learned fixed points, optimized for syntactic prediction and semantic coherence within a word embedding space. Moreover, Alice models entailment $P \models Q$, as well as abduction, given hypothesis Q find a parsimonious premise P to interpret and explain it. In this way, LLM Alice is a semiotic channel whose capacity is determined by the mutual information between message Q and interpretation P .

Alice LLM that result in profiles with higher overall mutual information are sought for (i) identification, certification and comparison of LLMs; (ii) prompt and context design; (iii) risk analysis and disinformation detection; and (iv) adaptive systems that can dynamically adjust their semiotic profile to audience and task, see [1]. This provides a rigorous and actionable framework for understanding, evaluating, and designing LLM-mediated communication between AI agents.

state machine Ecila($\mathcal{S}, \Theta'$) [4]

Let corpus $\mathcal{C} = \{t_1, \dots, t_m\}$ where token $t_i \in \mathbb{R}^p$

Define transition function $\mathcal{S}$ by an implicit RNN: [7]

$$h_0 = 0 \in \mathbb{R}^q \text{ and } k_0 = 0 \in \mathbb{R}^q$$

$$u_i = \begin{pmatrix} t_i \\ h_{i-1} \end{pmatrix} \in \mathbb{R}^r, \quad r = p+q, \quad i \in [m]$$

$$k_i = \varphi(Ak_i + Bu_i) \in \mathbb{R}^q, \quad A \text{ well-posed for } \varphi,$$

$$h_i = \mathcal{S}(h_{i-1}, t_i) \triangleq Ck_i + Du_i \in \mathbb{R}^q$$

Let $\Theta' = \{(W_1, b_1), (W_2, b_2)\}$ denote parameters

of FeedForward network: $\text{FFN}_{\Theta'}(h_i) = \text{ReLU}(h_i W_1 + b_1) W_2 + b_2$

Define output function $o'_i = \mathcal{O}'(h_i) \triangleq \text{FFN}_{\Theta'}(h_i)$

Ecila($\mathcal{S}, \Theta'$) training problem:

Learn matrices A, B, C, D and Fixed points k'_i , and FFN parameters

Θ' to minimize: $\sum_{i=1}^m 1 - \text{Cos}(o'_i, \text{Emb}(t_i))$

where $\text{Emb}: \mathbb{R}^p \rightarrow \mathbb{R}^{p'}$ is a word embedding and

$$\text{Cos}(a, b) \triangleq \frac{\langle a, b \rangle}{\|a\| \|b\|}.$$

state machine Alice($\mathcal{S}, \mathcal{O}$) [4,5]

From Ecila($\mathcal{S}, \mathcal{O}'$) we will use $\{h_i\}_{i=1}^m$,

$\{k_j\}_{j=1}^m$ and $\{v_j\}_{j=1}^m$ where $v_j = o'_j$

to define output $\mathcal{O}$ for Alice($\mathcal{S}, \mathcal{O}$) as follows:

Attention Layer [6]

Input h_i .

Let score be a similarity function as a kernel.

For $i \in [m]$, define probabilities

$$\alpha_{ij} = \underset{j \in [m]}{\text{softmax}} \text{ score}(h_i, k_j)$$

Let $\hat{h}_i = \sum_{j=1}^m \alpha_{ij} v_j$ and define update

$$h'_i = \text{FFN}_{\theta}(h'_i) \quad (\dim h'_i = \dim k_j)$$

Output h'_i

→

Stack L attention layers $\Theta = \{\theta_1, \dots, \theta_L\}$ composing updates

FFN_{θ_L} and now let h'_i denote the final L -th update.

After L layers, for $i \in [m]$ define probabilities

$$\beta_{ij} = \underset{j \in [m]}{\text{softmax score}}(h_i', k_j) \text{ and}$$

$$\text{let } j_i^\beta = \underset{j \in [m]}{\arg \max} \beta_{ij}$$

$$\text{Define token continuation } o_{i+1} = \Phi(h_i) \triangleq t_{j_i^\beta}$$

So attention layers construct output Φ .

Define the average Renyi α -entropy of $\{t_1, \dots, t_m\}$ to be

$$E(\{t_i\}_{i=1}^m) = \frac{1}{m} \sum_{i=1}^m H_\alpha^i$$

$$\text{where } H_\alpha^i = H_\alpha(\{\beta_{ij}\}_{j=1}^m) \text{ and } H_\alpha \text{ is Renyi } \alpha\text{-entropy.}$$

Alice $(\mathcal{S}, \Phi)$ training problem:

Let $\lambda > 0$, learn feedforward network parameters Θ

to minimize: $\text{Perplexity}(o_2, \dots, o_{m+1}) + \lambda E(\{t_i\}_{i=1}^m)$

Alice is a state machine LLM that combines an implicit RNN transition with an attention based output over learned fixed points, optimized for syntactic prediction and semantic coherence within a word embedding space.

Entailment [3]

Suppose $Alice(\$, \mathcal{O})$ processes two predicate token

strings, premise $P = \{p_1, \dots, p_r\}$ and hypothesis $Q = \{q_1, \dots, q_s\}$

as follows: $h_j^P = \$ (h_{j-1}^P, p_j)$, $j \in [r]$, $h_0^P \triangleq 0$

$$h_j^Q = \$ (h_{j-1}^Q, q_j), j \in [s], h_0^Q \triangleq 0$$

To obtain $h_P \triangleq h_r^P$ and $h_Q \triangleq h_s^Q$

Let h'_i denote the final update in a stack of attention layers

in output $\mathcal{O}(h_i)$ of $Alice(\$, \mathcal{O})$. We drop the token

index i unless necessary. As before $\beta_j(h) = \underset{j \in [m]}{\text{softmax score}}(h'_i, k_j)$

and let $\beta(h) = (\beta_1(h), \dots, \beta_m(h))$ be the distribution

of h over $\{k_j\}_{j=1}^m$

For $0 < \tau < 1$, define $F_\tau(h) = \{k_j \mid \beta_j(h) > \tau\}$

Define (transitive) entailment $P \models_\tau Q$ iff $F_\tau(h_Q) \subseteq F_\tau(h_P)$

Let k be small integer, define contradiction $P \models_k \neg Q$

iff $|F(h_P) \cap F(h_Q)| < k$.

On training triples (P, Q, y) where

$y \in \{\text{entail}_\varepsilon, \text{neutral}, \text{contradiction}_\kappa\}$

$$\text{learn } \text{Ent}(h_P, h_Q) = \begin{cases} \text{entail}_\varepsilon & \text{if } P \models_\varepsilon Q \\ \text{contradiction}_\kappa & \text{if } P \models_\kappa \neg Q \\ \text{neutral} & \text{otherwise} \end{cases}$$

to minimize loss $\mathcal{L} = \mathcal{L}_{\text{Perplexity}} + \lambda \mathcal{L}_{\text{Rensi}} + \rho \mathcal{L}_F$

where $\mathcal{L}_F$ is a loss function for Ent ,

and $\mathcal{L}_{\text{Perplexity}}$ and $\mathcal{L}_{\text{Rensi}}$ are loss functions for

$\text{Perplexity}(o_2, \dots, o_{m+1})$ and $E(\{t_i\}_{i=1}^m)$ average Rensi α -entropy

respectively. Hence Alice($\mathcal{S}, \mathbb{O}$) can learn entailment $\models$.

Abduction [3]

Given a hypothesis Q , we wish to find a premise P

such that $P \models Q$. Token string Q maybe autoregressively

continued by Alice($\mathcal{S}, \mathbb{O}$) to generate $Q, P_1, P_2, \dots, P_n, n \geq 1$,

where P_i denotes a continuation sentence.

Choose $0 < \tau < 1$, For each P_j check if $P_j \models_{\tau} Q$.

Suppose $\mathcal{P} = \{P_j \mid P_j \models_{\tau} Q\} \neq \emptyset$

let $m = \min_{P_j \in \mathcal{P}} |\mathbb{F}_{\tau}(h_{P_j})|$ and

let P_{j^*} be the first P_j with $|\mathbb{F}_{\tau}(h_{P_j})| = m$

Abduction

Given non-empty $\mathcal{P}$, choose $P_{j^*} \in \mathcal{P}$

to entail Q ; $P_{j^*} \models Q$ is a parsimonious explanation.

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Semiotics For LLM [1]

Let $\alpha \in (0, 1) \cup (1, \infty)$ and $X \sim P_X$

Rényi α -entropy $H_{\alpha}(X) \triangleq \frac{1}{1-\alpha} \log \sum_x P_X(x)^{\alpha}$

For $(X, Y) \sim P_{X,Y} = P_X P_{Y|X}$.

Arimoto mutual information $I_{\alpha}^A(X; Y) = H_{\alpha}(X) - H_{\alpha}^A(X|Y)$

where $H_{\alpha}^A(X|Y) \triangleq \frac{\alpha}{1-\alpha} \log \sum_y \left(\sum_x P_X(x)^{\alpha} P_{Y|X}(y|x)^{\alpha} \right)^{\frac{1}{\alpha}}$

To quantify the uncertainty of text, define semiotic breadth

$$S \triangleq H_\alpha(h_Q) \text{ where } h_Q \sim \beta(h_Q)$$

the Renyi α -entropy of message Q .

We say P is an interpretation of Q if $P \models Q$

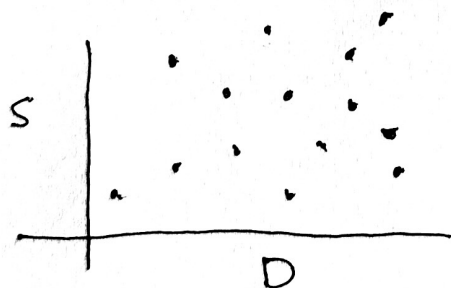
and define semiotic decipherability

$$D \triangleq I_\alpha^A(h_Q; h_P) = H_\alpha(h_P) - H_\alpha^A(h_P | h_Q) \quad [2]$$

where $P \models Q$, to quantify the reduction in uncertainty about an interpretation that results from observing a message.

To profile semiotic channel capacity $C_\alpha^A \triangleq \max_{h_Q} I_\alpha^A(h_Q; h_P)$

map S against D creating a cloud signature.



LLM semiotic signature

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